

EXPERIMENT 7

Aim

To build and train an **Artificial Neural Network (ANN)** using Keras/TensorFlow to predict whether a customer will churn based on given features.

Problem Statement

Customer churn prediction is important for businesses to retain customers.

The problem is to classify whether a customer will leave the bank or not using features like credit score, balance, age, and geography.

Dataset Link

<https://www.kaggle.com/datasets/shrutechlearn/churn-modelling>

Methodology

Data Collection & Cleaning

The dataset was loaded and unnecessary or irrelevant columns were removed to improve efficiency and reduce noise.

Data Preprocessing

Categorical features such as Gender and Geography were encoded into numerical values so they could be used by the neural network.

Data Splitting

The dataset was divided into training (80%) and testing (20%) sets to ensure proper evaluation of the model.

Feature Scaling

StandardScaler was applied to normalize the data, ensuring all features are on the same scale and improving model performance.

ANN Model Architecture

An Artificial Neural Network was built with an input layer, two hidden layers using ReLU activation, and an output layer using Sigmoid activation for binary classification.

Model Compilation

The model was compiled using the Adam optimizer for efficient learning and binary crossentropy as the loss function.

Model Training

The model was trained for 50 epochs, allowing it to learn patterns in the data through forward and backward propagation.

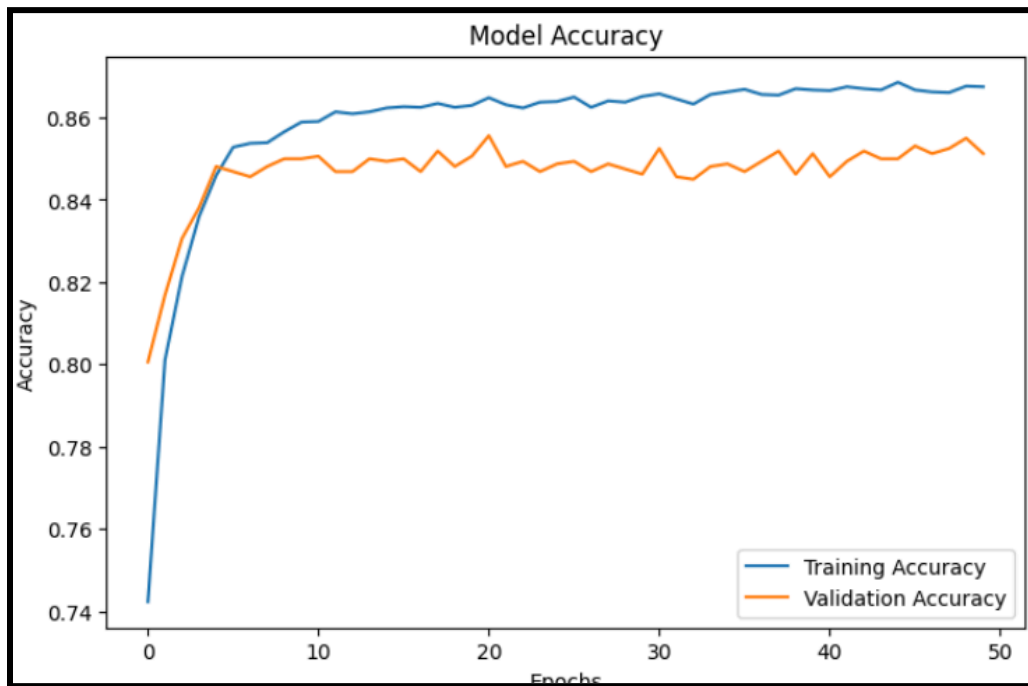
Prediction & Evaluation

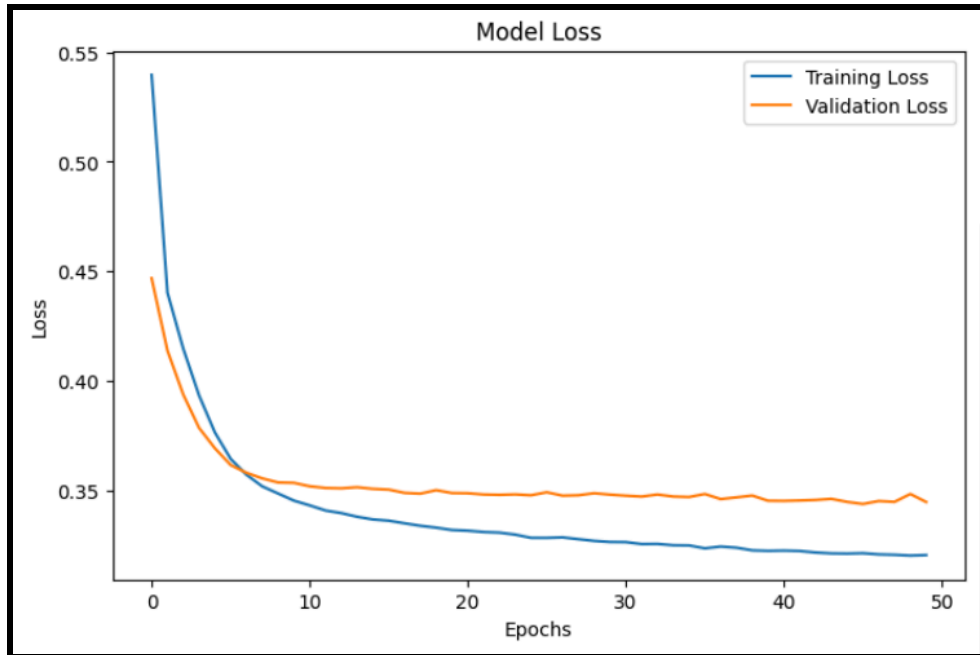
The trained model was evaluated using accuracy and a confusion matrix to measure performance and classification results.

Result

- **Accuracy:** ~85%
- **Confusion Matrix:** Shows correct and incorrect predictions
- **Classification Report:** Precision, Recall, F1-score

Model Graphs:





Conclusion

The ANN model effectively predicted customer churn with good accuracy by learning complex patterns and relationships within the data. By processing multiple features simultaneously, the model was able to identify subtle indicators that contribute to customer behavior and churn.

Feature scaling played a crucial role in improving performance, as it ensured that all input variables contributed equally during training and helped the model converge faster. This resulted in more stable and accurate predictions.

Overall, the model can be highly useful for businesses, as it helps in identifying customers who are likely to leave in advance. This enables companies to take proactive measures such as targeted marketing or retention strategies to reduce churn and improve customer satisfaction.